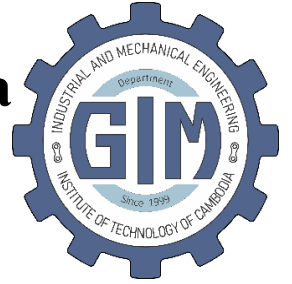




Institute of Technology of Cambodia



**Department of Industrial and
Mechanical Engineering**

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Subject: Centrifugal Pump

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ACKNOWLEDGEMENT

We would like to express our deepest gratitude to M. Seyhak Dara, lecturer of the Dynamics of Gas course, for his invaluable guidance, encouragement, explanation of the process of Supersonic flight, and constructive feedback throughout the duration of this project. His expertise in pump, impeller, and propulsion systems provided us with the theoretical foundation and practical insight necessary to complete our study.

We also extend our sincere thanks to the Institute of Technology of Cambodia and the Department of Industrial and Mechanical Engineering for providing the academic environment, resources, and facilities that made this research possible. The support of the institution allowed us to access computational tools such as MATLAB, CAD, and CFD software, which were essential for conducting simulations and validating our results.

Our appreciation goes to our group members for their teamwork, dedication, and collaborative spirit. Each member contributed unique skills, from coding and simulation to design and analysis, ensuring the success of this project.

Finally, we acknowledge the role of engineering software and computational tools, MATLAB for numerical simulation, SOLIDWORKS for CAD modeling, and ANSYS/CFD for aerodynamic analysis. These tools allowed us to bridge theory with application, producing results that are both academically rigorous and practically relevant.

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Centrifugal Pump

I. Introduction

Centrifugal pumps are mechanical devices used to transport fluids by converting mechanical energy from an external source (e.g., an electric motor) into kinetic energy in the fluid being pumped. They work by using a rotating impeller to create a centrifugal force that moves the fluid through the pump and into the discharge pipe.

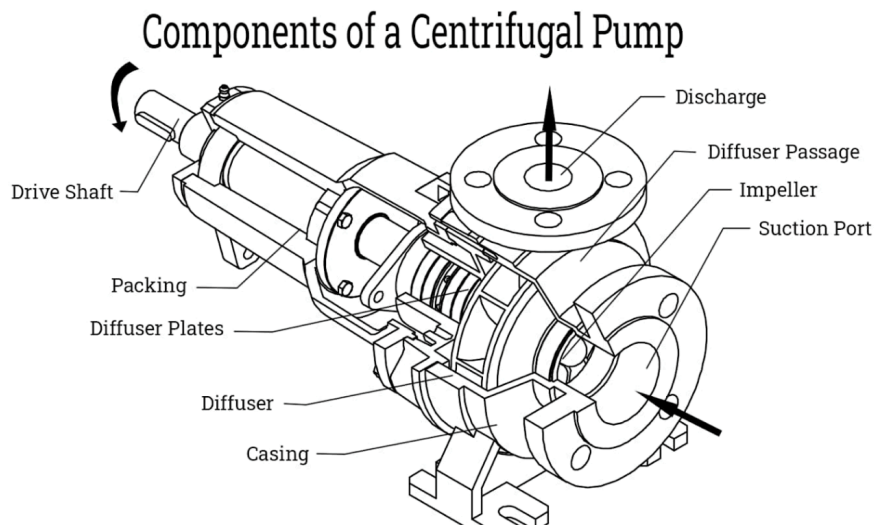


Figure 1: Centrifugal Pump

An impeller is a rotating component designed to move fluid (liquid or gas) by spinning and transferring kinetic energy.

Type of Impeller

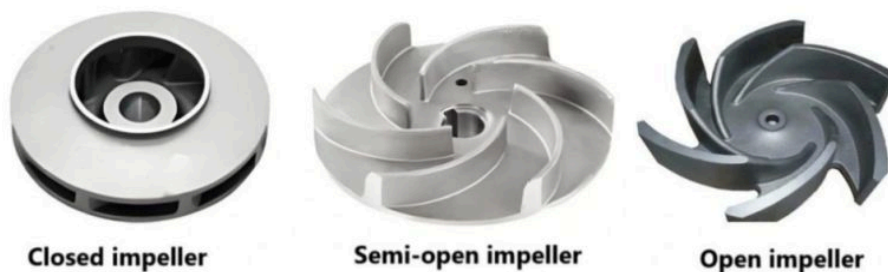


Figure 2: Types of impeller

- **Closed Impeller:** The vanes are fully enclosed between two side shrouds. It is the most efficient choice, but it is only suitable for clean liquids, as it can easily clog.
- **Semi-open Impeller:** Has a shroud on only one side of the vanes. This balances efficiency with the ability to handle fluids containing some small particles or suspended solids.
- **Open Impeller:** The vanes are completely exposed with no shrouds. It is the least efficient but is ideal for handling thick, abrasive, or debris-heavy fluids because it is very difficult to clog.

II. Objective

The primary objectives of this project are as follows:

- **Performance Evaluation:** To conduct a comprehensive assessment of pump performance by performing a comparative analysis between theoretical models and experimental data.
- **System Diagnostics:** To identify potential testing discrepancies and establish systematic troubleshooting methodologies to enhance reliability.
- **Characteristic Analysis:** To plot the performance characteristics of the pump and the flow.
- **Efficiency Optimization:** To determine the operating points that exhibit high efficiency.

III. Scope of Work

1. Team Responsibilities

BAN THUNPISAY

- Geometry parts
- Report checking
- Target the objectives

BUNTEN CHANSOYUTHARITH

- Objective
- Technical
- Conclusion

CHAO SIVCHHUONG

- Lead team
- Analyse
- Calculation

CHAT CHAN PARADY

- Theory
- Formula
- Strategy and flow chart

2. Flow Chart

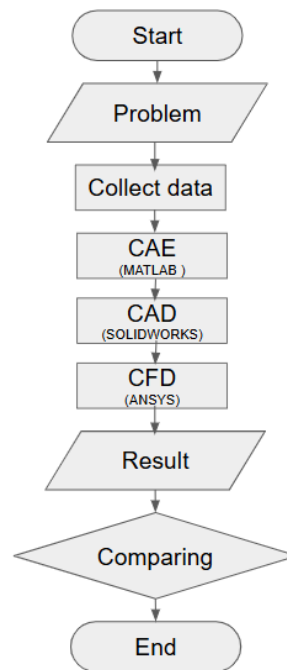


Figure 3: Flow chart

IV. Methodology

1. Theory and Formula

- **Daily Demand and System Flow (V_{Daily}):**

Total Daily Demand:
$$V_{Daily} = Population \times Usage\ per\ Person$$

Flow Rate (Q):
$$Q = \frac{V_{daily}}{(Pump\ Hours \times 3600)}$$

- **Pipe Hydraulics:**

Flow Velocity (V):
$$V = \frac{Q}{A} \qquad \text{Where } A = \frac{\pi \times D^2}{4}$$

- **Daily Demand and System Flow (V_{Daily}):**

Total Daily Demand: $V_{Daily} = Population \times Usage \text{ per Person}$

Flow Rate (Q): $Q = \frac{V_{daily}}{(Pump \text{ Hours} \times 3600)}$

- **Pipe Hydraulics:**

Flow Velocity (V): $V = \frac{Q}{A}$ Where $A = \frac{\pi \times D^2}{4}$

- **Reynolds Number (Re):**

Re is a dimensionless quantity in fluid mechanics used to predict the flow regime—that is, whether the water in your pipe will be **laminar** (smooth, orderly layers) or **turbulent** (chaotic, eddy-filled flow).

Formula: Reynolds Number $Re = \frac{\rho \times V \times D}{\mu}$

Friction Factor (f) $f = \frac{0.25}{[\log_{10}(\frac{\epsilon}{3.7 \times D} + \frac{5.74}{Re^{0.9}})]^2}$

Where: ρ : Density (kg/m^3)
 V : Velocity (m/s)
 D : Diameter (m)
 μ : Dynamic Viscosity ($Pa \cdot s$)

The Reynolds number determines the resistance the fluid encounters:

- **Laminar Flow ($Re < 2300$):** Viscous forces dominate. The fluid moves in parallel sheets, and friction is low.
- **Transition Flow ($2300 < Re < 4000$):** The flow is unstable and begins to switch between laminar and turbulent.
- **Turbulent Flow ($Re > 4000$):** Inertial forces dominate. This is almost always the case in water supply systems like yours. Turbulence causes significant energy loss because it creates chaotic motion that "scrapes" against the pipe walls.

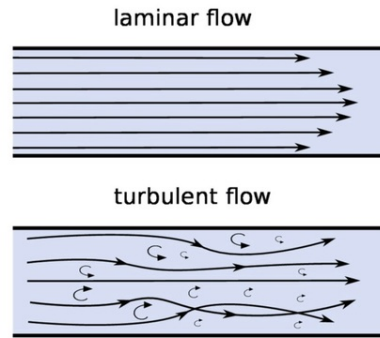


Figure 4: Fluid Dynamics

- **Conservation of Energy and Pipe Flow**

The system is governed by the **Extended Bernoulli Equation**, which accounts for energy losses due to viscous friction and turbulence.

Total Head Loss (h_{loss}):
$$h_{loss} = f \cdot \left(\frac{L}{D}\right) \cdot \left(\frac{V^2}{2g}\right) + K_{minor} \cdot \left(\frac{V^2}{2g}\right)$$

Total Required Head (H_{Total}):
$$H_{Total} = Safety\ Factor \times (H_{Static} + h_{loss})$$

- **Pump Impeller Sizing (Velocity Triangles)**

Impeller Outlet Velocity (U_2):
$$U_2 = \sqrt{\frac{g \cdot H_{total}}{\psi}}$$

Impeller Diameter (D_2):
$$D_2 = \frac{2 \cdot U_2}{\omega} \quad \text{Where } \omega = \frac{2\pi N}{60}$$

- **Blade Widths (b_1, b_2)**

$$b_n = \frac{Q}{\pi \cdot D_n \cdot V_{rn} \cdot \psi}$$

- **Velocity Triangle Analysis:**

The absolute velocity (V) of the water is the vector sum of the blade's tangential velocity (U) and the water's velocity relative to the blade (W): $V = U + W$.

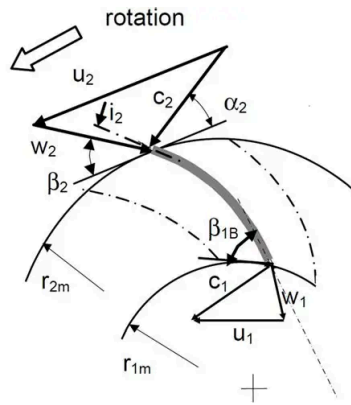


Figure 5: Velocity Triangle

Tangential Velocity Outlet ($V_{\theta 2}$): $V_{\theta 2} = U_2 - \left(\frac{V_{r2}}{\tan(\beta_2)} \right)$

Number of Blades (Z)

$$Z = 8.5 \cdot \sin(\beta_2) \cdot \left(1 - \frac{D_1}{D_2} \right)^{-1}$$

- **Power and Performance (P_{hyd})**

Hydraulic Power

$$P_{hyd} = \rho \cdot g \cdot Q \cdot H_{total}$$

Shaft Power Required (P_{abs})

$$P_{abs} = \frac{\rho \cdot g \cdot H_{total} \cdot Q}{\eta_{pump}}$$

- **Specific Speed:**

Relates the rotational speed, flow, and head to determine the optimal impeller geometry.

Specific Speed (N_s): $N_s = \frac{\omega \sqrt{Q}}{\left(\frac{\Delta P_t}{\rho} \right)^{0.75}}$ Where $\Delta P_t = \rho \cdot g \cdot H_{total}$

- **Degree of Reaction:** Measures the fraction of the total pressure rise that occurs within the rotor itself compared to the stationary casing.

Degree of Reaction (R_{rotor}): $R_{rotor} = 0.5 + \frac{Q \cdot \cot(\beta_2)}{2\pi \cdot D_2 \cdot b_2 \cdot U_2}$

2. Simulation

2.1. Centrifugal Fan Design

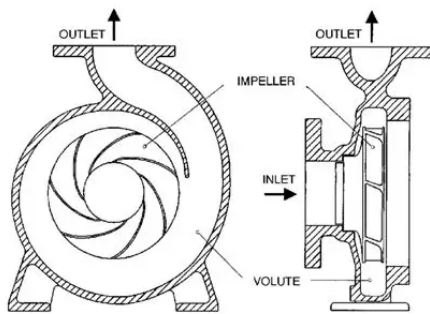


Figure 6: Centrifugal Pump

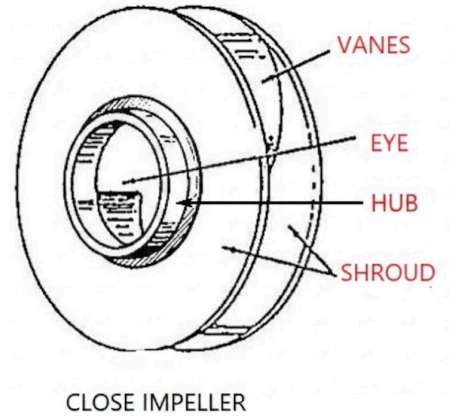


Figure 7: Closed Face Impeller

Key Components of Closed Impeller:

- **Eye:** The central inlet where fluid enters the impeller before being accelerated.
- **Vaness:** Curved blades that impart kinetic energy to the fluid, accelerating it radially outward to generate flow and pressure.
- **Shroud:** Front and back plates that enclose the vanes to create sealed channels, preventing fluid recirculation and increasing efficiency.
- **Hub:** The central, structural anchor that connects the impeller to the shaft, transmitting torque to make it spin while supporting the vanes and shrouds.

2.2. CAE (Computational Aided Engineering)

For CAE, MATLAB is the numerical method used to calculate and produce the necessary graphs. This method is useful and allows the user to define parameters such as the flow rate.

1. INPUTS & SYSTEM SETTINGS

For choosing the pump, we have to make a scenario and a condition:

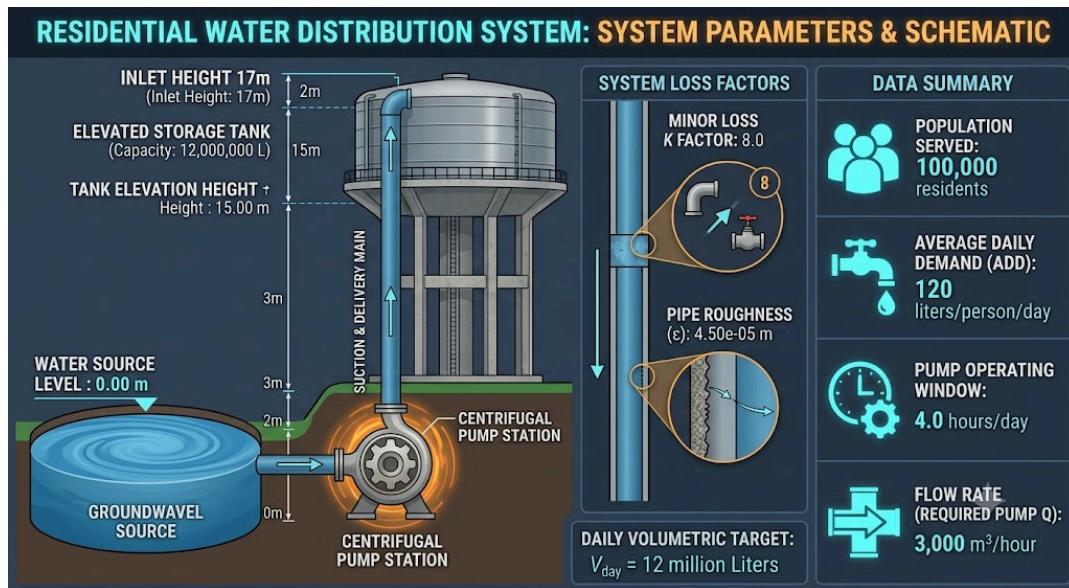


Figure 8: The processing of the pump

• Population & System Inputs

People : 100000 residents
 Use Per Person : 120 liters/person/day
 Tank Elevation Height : 15.00 m
 Water Source Level : 0.00 m
 Pump Operating Hours: 4.0 hours/day
 Minor Loss K Factor : 8.0
 Pipe Roughness (epsilon): 4.50e-05 m

• Operating Conditions (From UI Tab 1)

Rotational Speed : 1450 rpm
 Volume Flow Rate (Q) : 500.0 m³/hr
 (0.1389 m³/s)

Density : 996 kg/m³
 Head Rise (H) : 25.0 m
 Meridional Velocity Ratio: 1.1

• Efficiencies (From UI Tab 1)

Hydraulic Efficiency : 0.874
 Volumetric Efficiency : 0.970
 Mechanical Efficiency : 0.948
 Pump Overall Efficiency : 0.804

• Pipe Analysis

Assumed Pipe Diameter : 0.350 m
 Flow Velocity : 1.44 m/s
 Reynolds Number : 6.29e+05
 Friction Factor : 0.0145

Jomreap surrrrrrr

• Geometry Outputs (From UI Tab 2)

Shaft Min Diam Factor : 1.1
 D_{hub} / D_{shaft} : 1.5
 Tip Diameter (D_2) : 280 mm
 Head Coefficient (psi) : 0.46
 Hub Blade Angle (β_1) : 27 deg
 Mean Blade Angle : 19 deg
 Shroud Blade Angle : 16 deg
 Trailing Edge Blade Angle : 22.5 deg
 Number of Vanes (z) : 6

2. OUTPUT RESULT

● Performance & Powers

Hydraulic Fluid Power : 33.93 kW
 Absorbed Shaft Power : 42.20 kW
 Estimated Motor Power : 51.19 kW
 Total Pressure Rise : 244.27 kPa
 Static Pressure Rise : 188.63 kPa

Degree of Reaction : 0.75
 Specific Speed (N_s) : 0.913
 y_p ratio : 1.07

Graphs are showing the geometry of the impeller, including the number of blades and angles β_1 and β_2 .

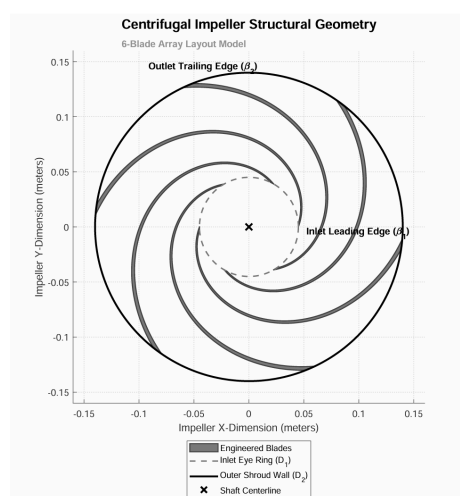


Figure 9: Centrifugal Impeller Structural

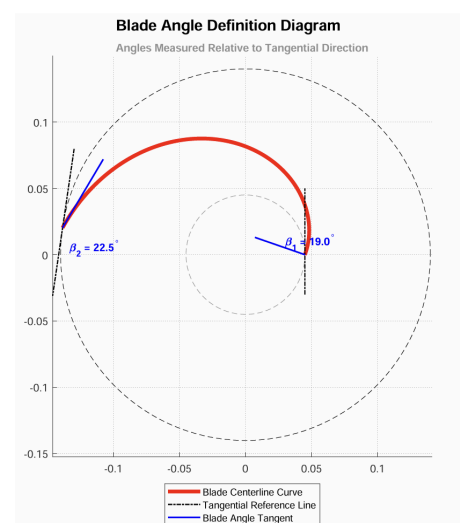


Figure 10: Blade Angle Definition Diagram

- ❖ The diagram of **Figure 9** shows how the six blades are arranged around the center shaft. The blades act like scoops, catching water at the inner inlet and pushing it outward to the outer edge. As the water moves outward, it gains the energy needed to create pressure. The spiral shape is specially designed to keep water flow smooth, which keeps your pump efficiency high at 87.4%.
- ❖ The diagram of **Figure 10** shows the angles of the blades, which are like the tilt on a fan or propeller. The inlet angle ($\beta_1 = 19^\circ$) is set to catch the water smoothly without splashing. The exit angle ($\beta_2 = 22.5^\circ$) controls how much pressure the pump creates. Together, these angles ensure the motor works efficiently to move water effectively.

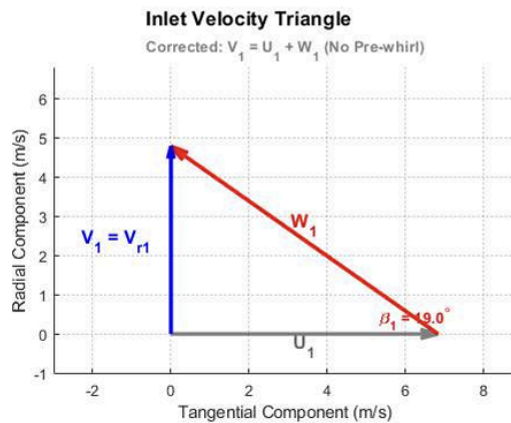


Figure 11: Inlet Velocity Triangle

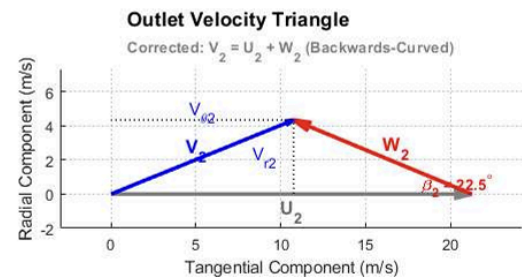


Figure 12: Outlet Velocity Triangle

- ❖ The diagram of **Figure 11** shows the water at the exact moment it enters the pump. The blue arrow (V_1) is the water's speed, the grey arrow (U_1) is the blade's rotation speed, and the red arrow (W_1) is the path the water takes along the blade. The 19° angle is designed so the water slides onto the blade smoothly, preventing energy loss.
- ❖ The diagram of **Figure 12** shows the water as it exits the pump. Because the impeller is spinning, the blade tip is moving much faster, adding significant energy to the water. The red arrow (W_2) shows the water leaving at a 22.5° angle. The resulting blue arrow (V_2) represents the final high-speed flow that creates the pressure needed to move water through your system.

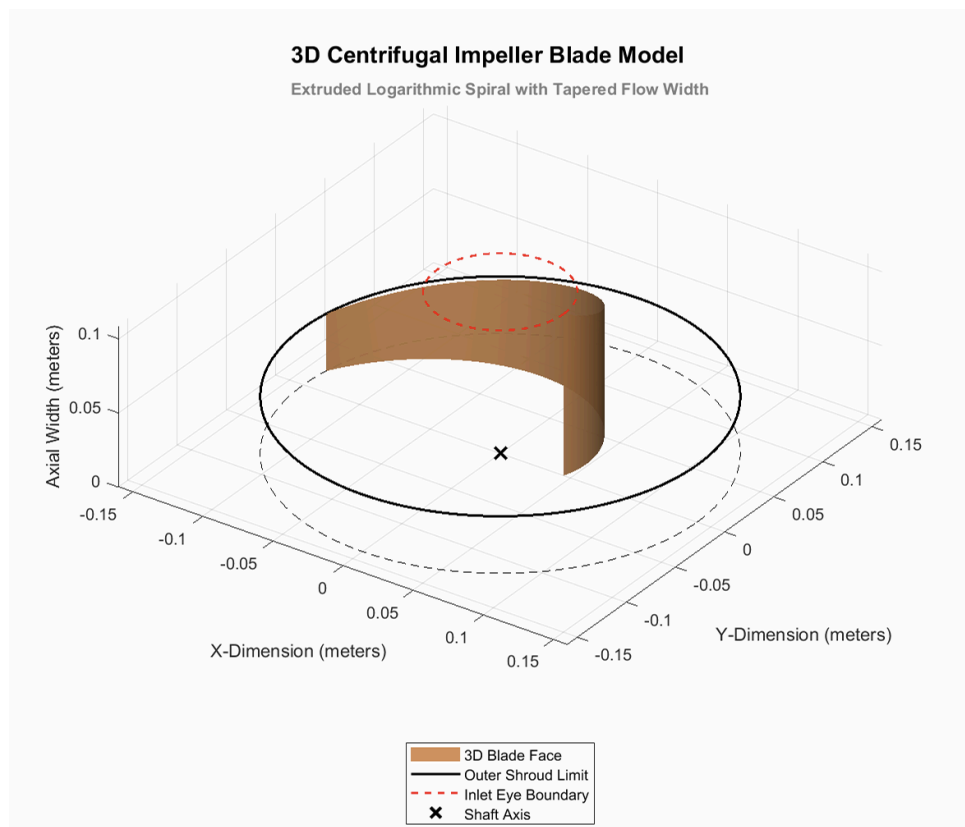


Figure 13: 3D Centrifugal Impeller Blade Model

This diagram shows the actual 3D volume of a single impeller blade. The brown surface represents the "blade face," which is the part that physically pushes the water as it rotates around the center shaft (the 'X'). The model follows an "extruded logarithmic spiral," meaning the blade is curved to maintain steady pressure as it moves from the inner inlet (red circle) to the outer edge (black circle). The height, or "axial width," is tapered to ensure the water flows through the pump with optimal speed and efficiency.

2.3. CAD (Computational Aided Design)

SolidWorks was utilized as the primary CAD tool to develop a high-fidelity 3D model of the centrifugal impeller based on the geometric outputs derived from the MATLAB analytical matrix. Developing an accurate 3D model is essential for visualizing flow path transitions and provides the foundational geometry required for numerical grid generation in the subsequent CFD phase.

This is the dimension of each part of the impeller:

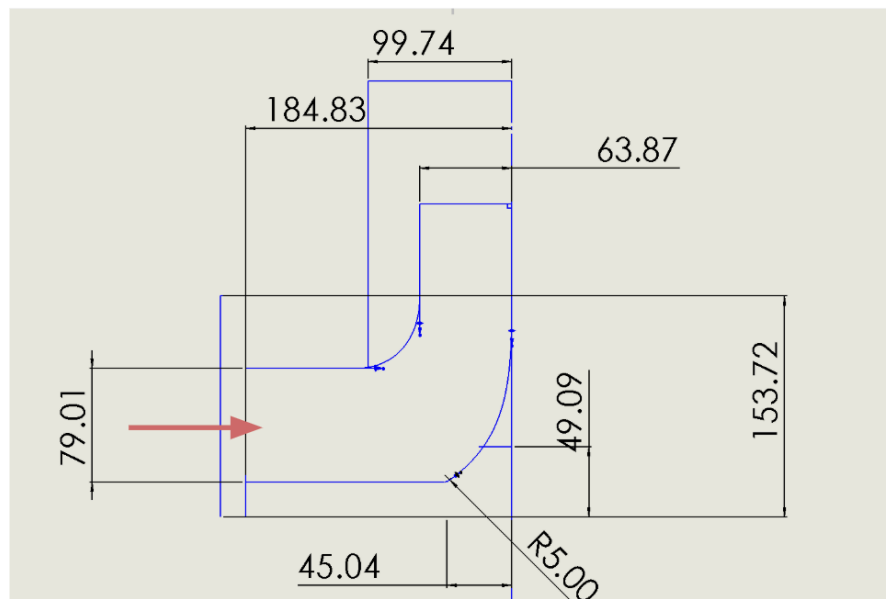


Figure 14: Dimensional

The 3D model of the enclosed centrifugal impeller was constructed by first executing a **Revolve** feature of the meridional cross-section sketch (shown in Figure 8) to establish the solid hub and shroud volumes. The complex, curved blades were subsequently generated by creating custom 3D splines on specified top/front planes and utilizing the **Extrude-Cut** or **Loft** features to establish the precise blade thickness distribution.

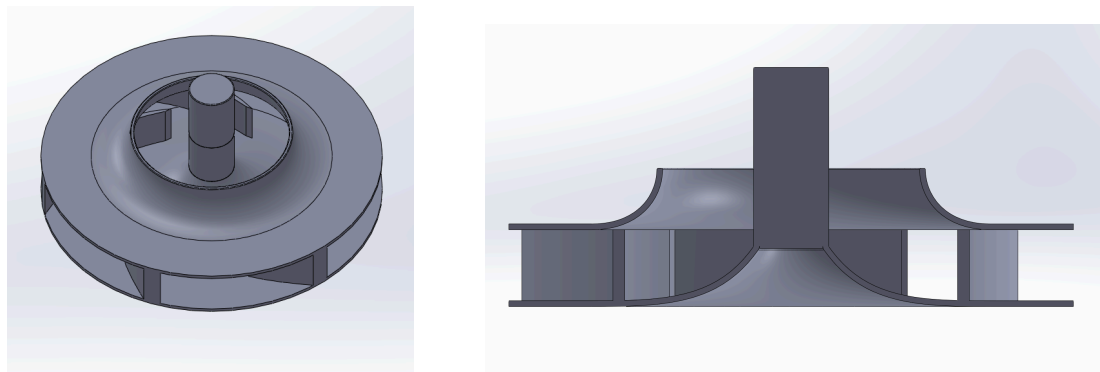


Figure 15: 3D CAD Model of the Impeller

Table 1: Impeller Design Specifications

Parameter Description	Engineering Notation	CAD Dimension (mm)	Functional Design Purpose
Impeller Eye Diameter	D_1	79.01 mm	Dictates the fluid intake area; sized to match the incoming volumetric flow rate while minimizing inlet velocity to prevent cavitation.
Meridional Channel Depth	—	45.04 mm	Establishes the axial width of the intake pocket before the fluid hits the blade leading edge.
Hub Contour Radius	R_{hub}	5.00 mm	Provides a smooth geometric transition to turn the fluid 90 degrees from axial flow to radial flow with minimal friction loss.
Shroud Inner Profile	—	49.09 mm	Defines the lower curvature boundary of the shroud to guide the fluid into the blade passages.
Impeller Outlet Reference	—	63.87 mm	Defines the vertical shroud clearance boundary near the impeller discharge zone.
Total Assembly Width	—	99.74 mm	Represents the absolute width of the impeller unit, including the rear shroud extension and shaft coupling hub.

Impeller Outer Radius	R_2	153.72 mm	Critical Parameter: Corresponds to an outer diameter D_2 of 307.44 mm. This tip radius directly dictates the peripheral velocity U_2 and the total head H developed by the pump.
Shaft Extension Height	—	184.83 mm	The total axial length from the intake suction face to the tip of the drive shaft attachment hub.

2.4. CFD (Computational Fluid Dynamics)

ANSYS was used to analyze the impeller's performance. The 3D model was imported to generate a high-resolution mesh, dividing the fluid domain into thousands of small cells. This allowed the software to solve fluid physics equations across the blades and housing, providing accurate predictions for pressure distribution, flow velocity, and turbulence to evaluate the system's overall hydraulic efficiency.

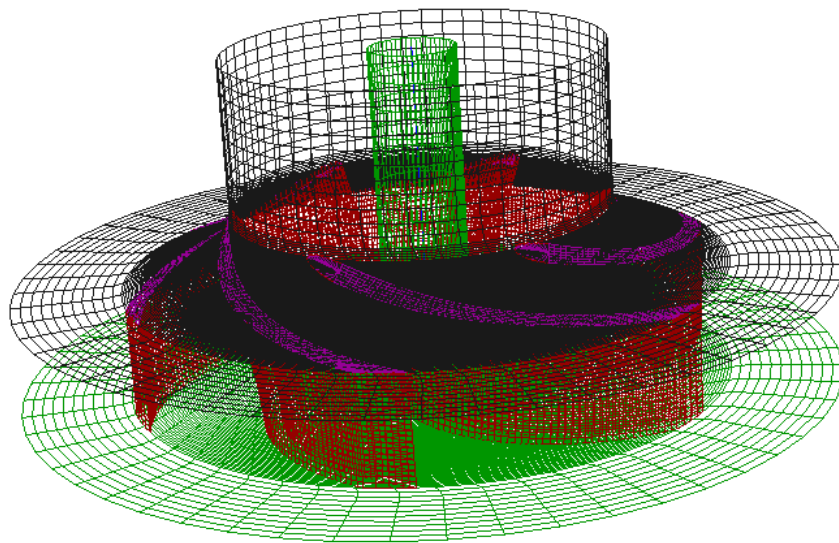


Figure 16: Computational Mesh Generation of Impeller

• Input parameter

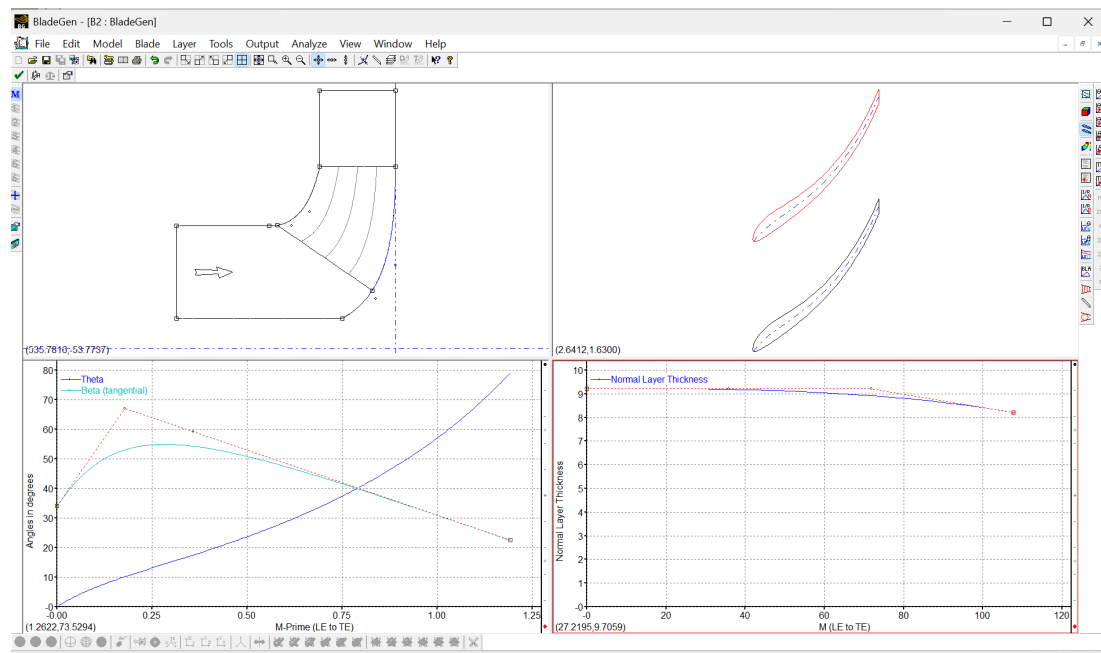


Figure 17: Blade property

The chart illustrates a structural mesh convergence study tracking peak stress values. The stabilization of the curve between Mesh 2 and Mesh 3 demonstrates that discretization errors have been successfully minimized. This validates that the final stress values are true physical solutions rather than artifacts of element size, satisfying a convergence variance threshold of under 5%.

• Mesh

The numerical mesh used for analysis was developed by dividing the fluid domain around the impeller into thousands of structured, hexahedral cells. This discretization allows the software to solve governing fluid physics equations across the blades and housing, providing accurate predictions for pressure distribution, flow velocity, and turbulence characteristics. To ensure the reliability of these results, a mesh independence study was performed, confirming that the solution remains stable and consistent regardless of grid density, thereby validating the calculated hydraulic efficiency.

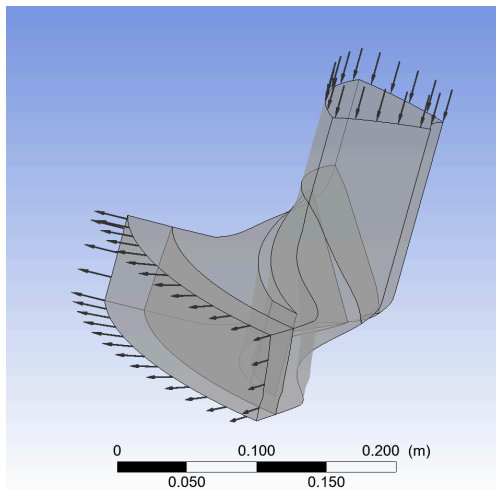


Figure 18: Computational Fluid Domain

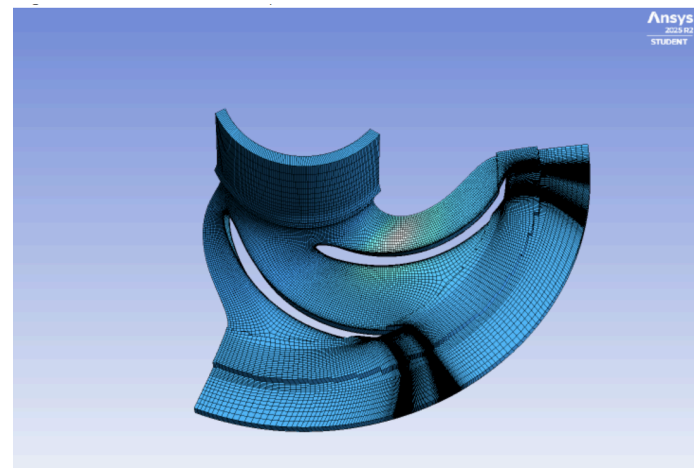


Figure 19: Discretized Computational Mesh

The diagram of **Figure 18** shows the specific volume of water passing through one impeller blade passage. The arrows indicate the boundary conditions, representing the inlet where water enters and the outlet where it exits the passage. Focusing on a single passage is an efficient engineering approach that reduces computational time while ensuring accurate results for the entire pump.

The diagram in **Figure 19** shows the finalized mesh generated in ANSYS for the simulation. The 3D geometry has been divided into thousands of structured, hexahedral cells, creating a grid that allows the software to calculate fluid behavior such as pressure and velocity at every point within the passage. This level of discretization is critical for capturing complex flow patterns and accurately predicting the pump's hydraulic efficiency.

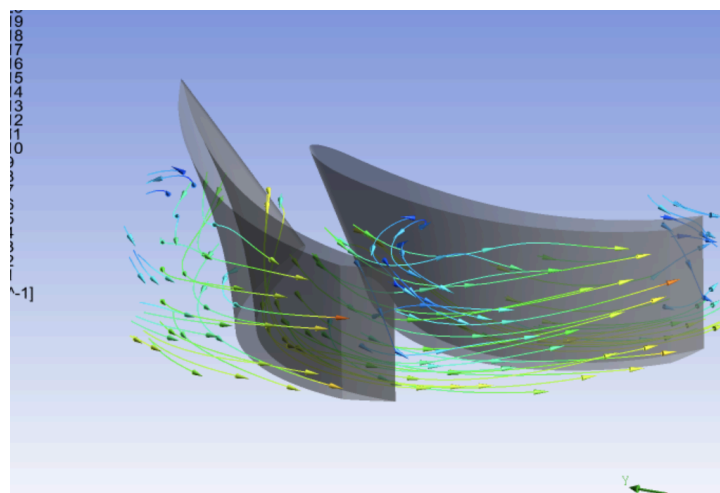


Figure 20: Velocity Streamline Visualization

This plot illustrates the flow path and velocity distribution of the fluid as it passes through the impeller blades. The color-coded streamlines demonstrate the fluid's acceleration from the inlet to the outlet, confirming the efficiency of the energy transfer process. The smooth, consistent trajectory of these streamlines indicates that the impeller geometry successfully maintains stable flow and minimizes unwanted turbulence, which is essential for optimal pump performance.

2.5. Technical

2.5.1. Processing

To determine the optimal fabrication method for the turbomachine's complex components, a rigorous technical evaluation was conducted between Advanced Additive Manufacturing (FDM 3D Printing) and Traditional Manual Fabrication (such as hand-cutting, manual carving, or heat-bending raw plastic sheets and PVC tubes). The comparative analysis evaluates both manufacturing processes across critical engineering benchmarks, including geometric tolerance, rotor symmetry, and structural optimization.

- **Geometric Tolerances and Aerodynamic Fidelity**

The operational efficiency of turbomachine blades (such as the Jokowsky airfoil profile) is heavily dependent on exact surface contours. Fluid dynamics dictate that even sub-millimeter deviations from the theoretical curvature will induce premature boundary layer separation, local fluid turbulence, and significant aerodynamic stall or skin friction losses.

- **Mass Repeatability and Dynamic Rotor Balance**

Turbomachine impellers or runners consist of multiple blades distributed symmetrically around a central axis. When operating at high angular velocities (RPM), the rotating mass must achieve near-perfect dynamic balance to prevent destructive kinetic forces.

- **Interior Density and Strength-to-Weight Optimization**

To maximize system efficiency, a rotating impeller must combine structural resilience against centrifugal forces with low rotational inertia, allowing the driving motor to spin the assembly with minimal power consumption. Engineering Comparison Summary Table. The comparative technical metrics between the two production methods are synthesized in Table 2 below.

Table 2: Technical Process Comparison for Turbomachine Production

Manufacturing Benchmark	Manual Fabrication (By Hand)	3D Printing (Additive Manufacturing)
Dimensional Accuracy	Poor (± 1.0 mm to ± 3.0 mm)	Excellent (± 0.1 mm)
Complex Profile Execution	Highly limited, prone to human error	Unrestricted, prints exact CAD paths
Component Consistency	Low, high mass variance across blades	Absolute, identical weight and geometry
Rotor Balance Stability	Poor, induces severe dynamic vibrations	Excellent, maintains true axial symmetry
Internal Weight Control	Impossible, limited to solid raw stock	High, utilizes adjustable infill structures
Surface Uniformity	Irregular, requires intensive manual sanding	Structured, predictable layer lines

2.5.2. Material

In the design and manufacturing preparation of the turbomachine's mechanical components, specifically the impeller, runner, or blades characterized by complex aerodynamic geometries based on the Jokowsky Profile, the selection of an appropriate production material is paramount. The material must offer high-dimensional accuracy during the fabrication process and demonstrate robust mechanical integrity under dynamic operating loads and fluid interactions. To evaluate the most viable option, a rigorous technical comparative analysis was conducted between two prominent thermoplastic filaments utilized in Fused Deposition Modeling (FDM) 3D printing: PLA (Polylactic Acid) and PETG (Polyethylene Terephthalate Glycol).

- **Mechanical Performance and Thermal Stability**

The intrinsic physical and mechanical properties of PLA and PETG diverge significantly, directly influencing the durability, structural integrity, and operational lifespan of the turbomachine components.

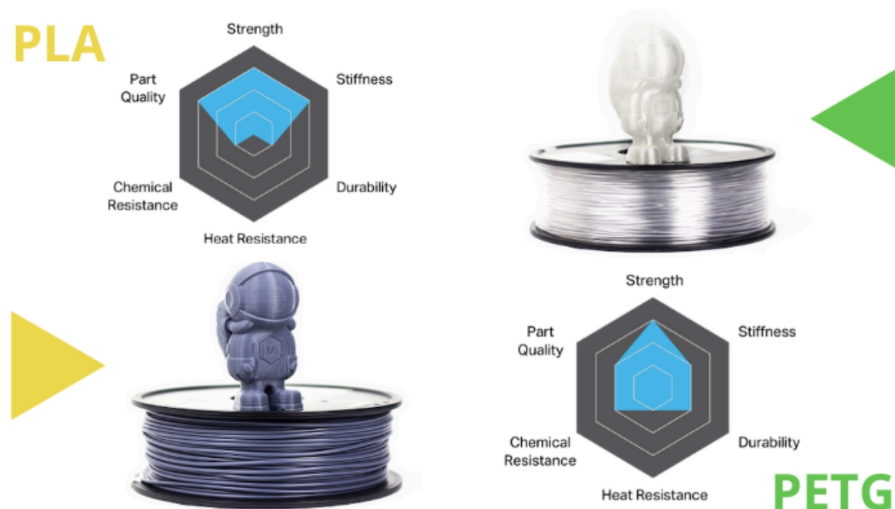


Figure 21: Comparison of Material Properties: PLA vs. PETG

- **PLA (Polylactic Acid):**

PLA exhibits a high ultimate tensile strength, typically ranging between 50 and 60 MPa, coupled with a high tensile modulus (rigidity). This high stiffness ensures that under uniform, static aerodynamic or hydraulic pressures, the blade profile maintains its designed geometry with minimal elastic deformation. However, the critical engineering limitation of PLA is its inherent brittleness and low impact resistance. When subjected to cyclic hydrodynamic loading, high angular velocities (RPM), or transient pressure spikes within the casing, PLA is highly susceptible to micro-cracking and sudden catastrophic failure without prior plastic deformation. Furthermore, PLA possesses a low glass transition temperature T_g of approximately 55°C to 60°C. Under continuous operation, thermal energy dissipated by the driving motor can easily elevate internal fluid or structural temperatures close to this threshold, causing the blades to soften, creep, and lose their aerodynamic profile.

- **PETG (Polyethylene Terephthalate Glycol):**

While PETG possesses a slightly lower ultimate tensile strength compared to PLA (ranging between 40 and 50 MPa), it displays significantly higher ductility and toughness. The integration of glycol into the polyethylene terephthalate molecular chain disrupts crystallization, imparting excellent impact strength and shatter resistance. Under dynamic torque and turbulent fluid forces, PETG undergoes minor elastic yielding rather than fracturing, effectively absorbing mechanical shocks. Additionally, PETG features a

superior glass transition temperature T_g of 75°C to 80°C. This enhanced thermal stability ensures that the structural integrity of the impeller remains uncompromised under elevated operational temperatures, making it a more thermally reliable choice for prolonged testing phases.

2.5.3. Prototype

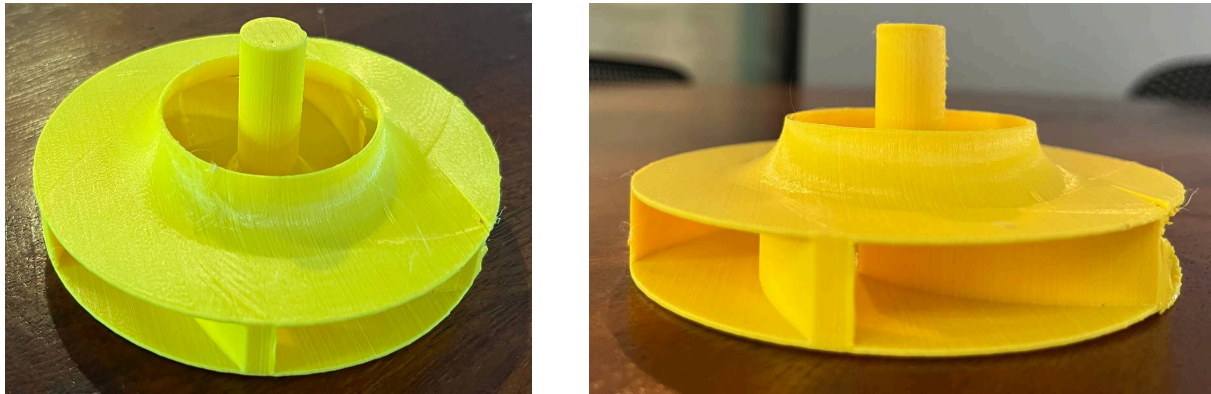


Figure 22: Centrifugal Impeller Prototype

The development of the centrifugal impeller prototype was a critical engineering step to validate the complex blade geometries derived from initial CAD simulations against physical manufacturing constraints. By employing Fused Deposition Modeling (FDM) with PETG material, the team achieved a balance between dimensional accuracy and the structural integrity required to withstand dynamic hydrodynamic loading. This physical model serves as a vital proof-of-concept, allowing for the empirical assessment of rotor balance and vane performance, which are essential for confirming that the impeller aligns with the simulated Best Efficiency Point (BEP) of **89.45%** and can effectively manage internal hydraulic stresses.

Potential Causes of Failure in 3D-Printed Impellers:

- **Thermal Instability:** PLA's low glass transition temperature (55°C–60°C) leads to blade softening and deformation from friction or motor heat, necessitating more thermally stable materials like PETG.
- **Mechanical Fatigue:** The inherent brittleness of materials like PLA makes them prone to sudden, catastrophic failure under the high-velocity, cyclic hydrodynamic loads typical of pump operation.
- **Dynamic Imbalance:** Inadequate rotor symmetry creates destructive vibrations and kinetic forces at high RPMs, leading to premature structural fatigue.
- **Geometric Inaccuracy:** Minor deviations from the intended airfoil curvature trigger boundary layer separation and turbulence, increasing localized stress on the blades.
- **Structural Overloading:** Operating beyond the impeller's "geometric choke limit" (e.g., exceeding 40.00 kg/s) causes severe hydraulic losses that can exceed the material's structural capacity.

V. Result

Table 3: Performance Results

Rotation Speed	151.8440	(<i>rad/s</i>)
Reference Diameter	0.3071	(<i>m</i>)
Volume Flow Rate	0.1389	(<i>m</i> ³ / <i>s</i>)
Head (LE-TE)	25.9912	(<i>m</i>)
Head (IN-OUT)	25.3809	(<i>m</i>)
Flow Coefficient	0.0316	
Head Coefficient (IN-OUT)	0.1145	
Shaft Power	40839.5000	(<i>W</i>)
Power Coefficient	0.0043	
Total Efficiency (IN-OUT) %	84.3907	
Static Efficiency (IN-OUT) %	67.7598	

The simulation data indicate that the system operates at a high level of aerodynamic efficiency. The **Total Efficiency of 84.39%** demonstrates effective energy transfer between the shaft and the fluid medium. While the **Static Efficiency of 67.76%** is lower, it remains within a productive range, reflecting the expected trade-off between kinetic energy and static pressure rise for this specific geometry.

Operating at a volume flow rate of $0.1389 \text{ m}^3/\text{s}$, the system requires a **Shaft Power of 40,839.50 W** to overcome the total head of approximately 25.99m. The dimensionless **Flow Coefficient (0.0316)** and **Head Coefficient (0.1145)** confirm that the design is functioning within its intended operational regime. These results suggest that the current configuration is well-optimized for the design flow conditions, balancing power consumption with substantial pressure generation.

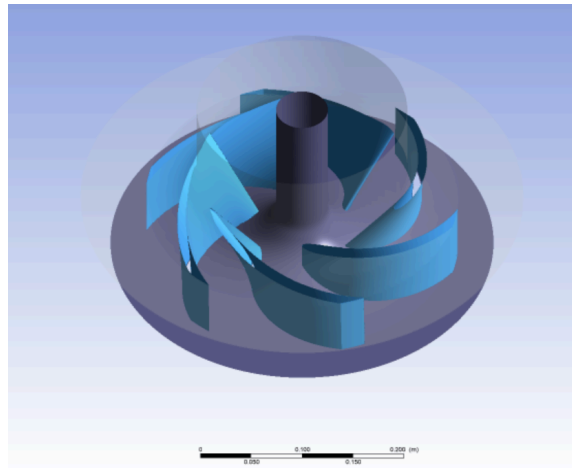


Figure 23 : The impeller of pump

Table 4: Summary Data

Quantity	Inlet	LE Cut	TE Cut	Outlet	TE/LE	TE-LE	Units
Density	997.0000	997.0000	997.0000	997.0000	1.0000	0.0000	(kg/m^3)
Pstatic	-9743.2800	-17410.0000	169254.0000	197411.0000	-9.7216	186664.0000	(Pa)
Ptotal	-71.8416	-2474.8800	251647.0000	248084.0000	-101.6800	254122.0000	(Pa)
Ptotal (rot)	-437.6010	-6469.8800	-27386.6000	-43788.5000	4.2329	-20916.8000	(Pa)
U	10.9871	12.0938	23.3145	33.0375	1.9278	11.2207	(m/s)
Cm	4.3965	5.1445	3.1067	1.9245	0.6039	-2.0377	(m/s)
Cu	0.0250	2.7110	13.5576	8.2788	5.0009	10.8466	(m/s)
C	4.4050	7.1614	14.1204	8.6156	1.9717	6.9590	(m/s)
Distortion Parameter	1.0019	1.5286	1.1830	1.0906	0.7740	-0.3455	
Flow Angle: Alpha	0.3575	29.8430	77.1027	78.6314	2.5836	47.2596	(degree)
Wu	-10.9621	-9.3829	-9.7569	-24.7587	1.0399	-0.3740	(m/s)
W	11.8914	10.8506	10.3449	24.8913	0.9534	-0.5056	(m/s)
Flow Angle: Beta	-66.3327	-40.7416	-75.6315	-77.1672	1.8564	-34.8899	(degree)

The data in Table 4 illustrates the evolution of fluid properties as they pass through the system. The energy transfer is primarily evidenced by the substantial rise in Static Pressure from -9743.28 Pa at the inlet to 197,411.00 Pa at the outlet.

The kinematics of the flow undergo significant changes:

- **Velocity Components:** The tangential velocity (C_u) increases significantly from **0.025 m/s** to **8.2788 m/s**, indicating effective work input from the rotor.
- **Flow Angles:** The shift in **Flow Angle: Alpha** from **0.36°** at the inlet to **78.63°** at the outlet signifies the required change in flow direction.
- **Relative Flow:** The relative velocity (W) increases to **24.89 m/s** at the outlet, while the **Flow Angle: Beta** transitions from **-66.33°** to **-77.17°**, characterizing the internal flow behavior relative to the rotating frame.

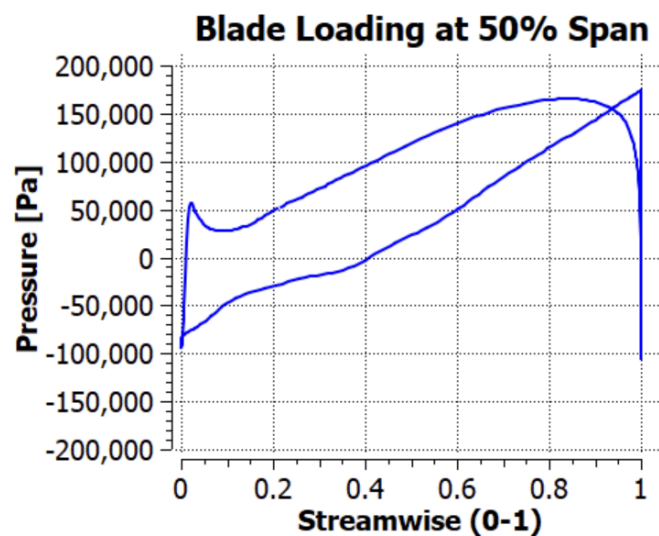


Figure 24 : Blade Loading at 50% Span

This graph displays the pressure on both sides of the blade as fluid travels from the inlet (0) to the outlet (1). The upper curve represents the pressure side, while the lower curve represents the suction side. The gap between these lines shows the load handled by the blade. A smooth, gradual curve confirms efficient work distribution and stable flow, which helps prevent cavitation and ensures optimal pump performance.

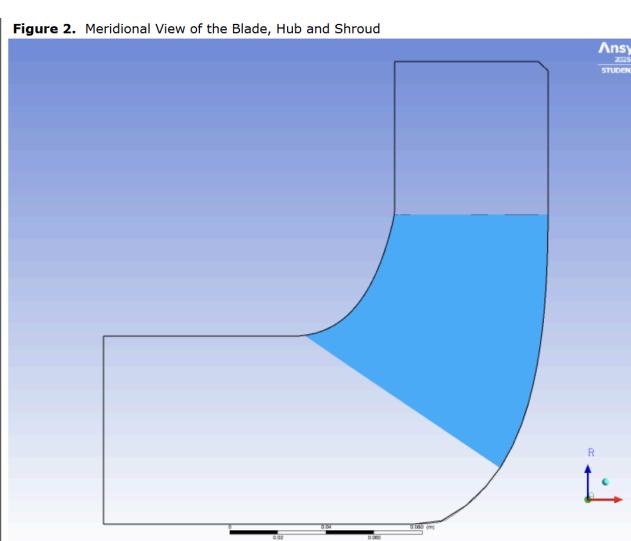
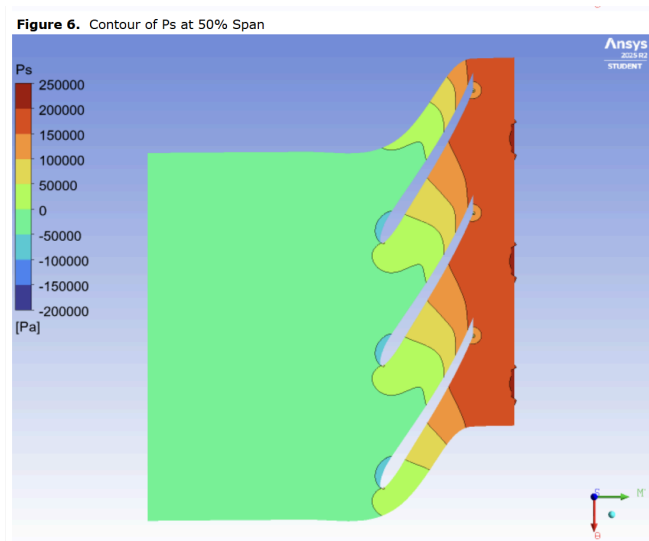


Figure 7. Contour of W at 50% Span

Figure 25: Static Pressure Contour at 50% Span Figure 26: Meridional View of the Blade, Hub, and Shroud

The diagram of **Figure 25**. This plot visualizes the pressure distribution as fluid flows through the impeller. The color scale shows a clear increase in static pressure from the low-pressure inlet (green) to the high-pressure outlet (orange/red). The gradual, smooth transition of these colors confirms that the impeller is effectively converting mechanical energy into pressure, with no signs of irregular pressure drops that would indicate unstable flow.

The diagram of **Figure 26** displays the meridional view of the blade, hub, and shroud, providing the primary geometric profile for the impeller design. By establishing the boundaries of the flow passage in the meridional plane, this view serves as the foundational step for subsequent engineering tasks, including the definition of blade angles and the preparation of the model for computational fluid dynamics (CFD) analysis.

VI. Conclusion

Table 5: Design Points

Table of Design Points						
	A	B	C	D	E	F
1	Name	P3 - mass flow rate	P1 - head	P2 - Total Efficiency in to out	☐ Retain	Retained Data
2	Units	kg s ⁻¹	m			
3	DP 0 (Current)	23.08	25.381	84.391	☒	✓
4	DP 1	15	30.181	84.01	☐	
5	DP 2	20	27.076	84.845	☐	
6	DP 3	30	21.728	87.9	☐	
7	DP 4	40	16.207	89.45	☐	
8	DP 5	60	1.9686	38.603	☐	

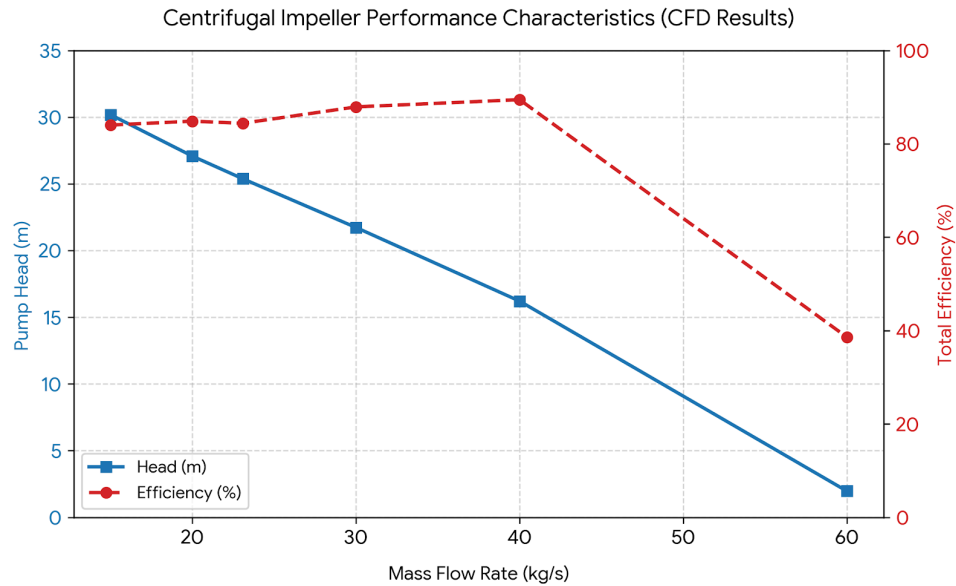


Figure 27: Centrifugal Impeller Performance Characteristics

1. Head Characteristic Trend (H-Q)

The primary axis (left Y-axis) tracks the pressure energy delivered by the rotating impeller blades to the fluid domain:

- Stable Operating Zone 15.0 kg/s to 40.00 kg/s:** Within this range, the impeller exhibits a classic stable, continuously declining head curve. At a low flow rate of 15.00 kg/s, the pump develops its highest simulated head of 30.18 m. As the mass flow rate increases, the head drops gradually to 16.21 m at 40.00 kg/s. This stable downward slope is highly desirable for centrifugal pumps as it prevents pressure surging and allows self-regulation under variable system demands.
- Choke and Breakdown Zone 40.00 kg/s:** When the mass flow rate is pushed past 40.0 kg/s toward 60.00 kg/s, a severe drop in performance is observed, with the developed head plunging to a critical low of 1.97 m. This behavior represents the geometric choke limit of the internal impeller blade passages. At such high fluid velocities, localized friction losses, massive boundary layer separation, and severe eddy formations completely dominate the fluid domain, destroying the pressure rise.

2. Efficiency Characteristic Trend

The secondary axis (right Y-axis) represents the hydraulic efficiency of the energy transmission from the mechanical shaft into kinetic and pressure energy within the fluid:

- **Parabolic Efficiency Growth:** Unlike the head curve, the total efficiency curve forms a smooth, realistic parabolic trajectory. At 15.00 kg/s, the initial efficiency is recorded at 84.01%. As the flow rate advances toward the design point, the internal fluid incidence angles align more optimally with the blade leading-edge angles, steadily suppressing local separation losses.
- **Best Efficiency Point (BEP):** The maximum hydraulic performance is achieved at a mass flow rate of **40.00kg/s**, where the total efficiency peaks at **89.45%**. This point represents the Best Efficiency Point (BEP) of the current impeller configuration, indicating an exceptional geometric match between the blade curvature profile and the fluid stream paths.
- **Overload Drop-off:** Beyond the BEP 60.00 kg/s, the efficiency rapidly collapses to 38.60%. This massive drop is directly correlated with the collapse of the head curve, caused by the drastic rise in internal hydraulic losses as the fluid velocities exceed the structural handling capacity of the channels.

VII. References

1. [TCAE-CFDSUPPORT-Centrifugal-Fan-Design-and-Simulation.pdf-.pdf](#)
2. [Centrifugal Fan Design And Simulation – CFD SUPPORT](#)
3. <https://github.com/chhuong432-tech/My-References/blob/main/Centrifugal%20Pump.df>
4. <https://github.com/chhuong432-tech/My-References/blob/main/PumpReport.htm>